

Quantifying and Measuring Confirmation Bias in Information Retrieval Using Sensors

Kaixin Ji

kaixin.ji@student.rmit.edu.au
RMIT University
Melbourne, Australia

ABSTRACT

It is well-known that cognitive bias influences information retrievers to receive information fairly. Among all the biases, we focus on confirmation bias, which is the most impactful and sometimes leads to polarization. Numerous projects have been working on improving search systems to provide fair results, while little work has been done on enhancing fairness from the user's side. Thus, this project aims to investigate a quantifiable approach to make the systems aware that retrievers carry confirmation bias and perform biased behaviors. Besides, some works have attempted to detect confirmation bias using web-logging or eye-tracking but failed to find differences between the search behaviors. In this regard, other quantifiable and objective measures should be applied. This project aims to collect multi-modal behavioral and physiological data using wearable sensors, and used as the input for machine learning techniques and build a bias-aware model.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; • **Information systems** → **Users and interactive retrieval**.

KEYWORDS

confirmation bias, information processing activities, physiological signals, information seeking, wearable sensors

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1 INTRODUCTION

Information seeking is the process in which people search and access information with the highest perceived trustworthiness and build mental models by linking and diverting one factor to the other [15]. Any factors would influence the current and future process [3],

e.g., the cognitive bias. Among all cognitive biases, Confirmation Bias (CB) has significantly influenced fair information receiving. Specifically, fair information receiving is to have balance views, equal trust, and cautious evaluation on any stances (assume all from credible sources) when user search for decision-making. The CB occurs when the users have prior beliefs and the presented information contradicts those beliefs [7, 10, 13, 15, 22, 31]. It causes cognitive dissonance and then evokes psychological discomfort, such as anger or anxiety [9]. To alleviate this discomfort, the users tend to search for information that aligns with their existing beliefs (selective exposure) [16, 35] or avoid those that do not match their existing belief (selective avoidance) [7]. However, detecting and measuring this discomfort can be difficult as it is not always noticeable.

Information Retrieval (IR) takes a user-input query, then provides a list of relevant search results based on the Probability Ranking Principle [28]. Users have the option to choose and view search results that may be influenced by bias, or they can enter a specific search query to obtain biased results [1]. Existing approaches usually access CB with subjective surveys or web-logging data [30, 35]. These methods are controllable by the users, which can be deceptive and lack accuracy. In this regard, physiological data can potentially provide an objective measurement and quantification of the CB and translate it into a format the computer system can understand [8, 27, 32–34]. Once the system can accurately quantify CB, there are two potential applications. The first application integrates into the system, detects users' bias, and prompts a warning message [6, 25] or other mitigation strategies. The second application uses this as an additional evaluation metric for developing the search system and minimizing occurrences of CB [5, 6]. Besides, with advances in wearable devices, most physiological signals can be captured with consumer-level devices, including Electrodermal Activity (EDA), Blood Volume Pulse (BVP), Heart Rate Variability (HRV), gaze movement, pupil responses, and physical movement.

This project aims to quantify and measure confirmation bias during information retrieval using sensor data. The overall research question is divided into two phases, four studies. Concerning the high sensitivity of physiological signals, in the first phase, we aim to gain a general view of the signals and investigate whether the signals should be treated respectively according to the interaction modalities. The first study is conducted in a general scenario beyond information search. Then, the second study continues with the aim but specifically focuses on the information search context. No bias involved so far. In the second phase, because CB has two presentations, viewing the supporting information more or perceiving the supporting information as more trustworthy [12], we look at these two presentations separately. The third study focuses

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only on the information evaluation stage (the latter presentation). Then, the fourth study extends to a more comprehensive setting, i.e., an information-seeking iteration from search to evaluation (both presentations).

2 RELATED LITERATURE

2.1 Detecting Confirmation Bias

Few projects have attempted to measure confirmation bias, and all highlighted that the CB is topic-sensitive [29]. Kessler and Zillich [12] investigated cognitive factors that occurred during online search with eye-tracking data. Their work focused on information about vaccination. The user study was conducted by collecting eye-tracking data while the participants were freely Internet-surfing after reading or not stimulus articles. Online behavior data were extracted as search queries, viewed-only results and selected results. Suzuki and Yamamoto [30] conducted the experiment with healthy food. The prior beliefs of participants are manipulated by presenting different background information. The results showed that people with high health literacy were more likely to view top-ranked results, as means of they are more likely to have CB, thus less scrutinizing the result pages. Besides, people with CB were less likely to change their beliefs despite increasing search sessions. Zillich and Guenther [35] claimed that eye blinks can refer to selective exposure as it impacts the visual information stream. They used a similar setting as Suzuki and Yamamoto [30] but a different topic, self-driving cars. However, all three studies found that the participants tended to perform neutral search behaviors by using neutral search terms, selecting and viewing neutral articles, and spending equal time on both stances. Suzuki and Yamamoto [30] argued that the selected topic could not trigger enough cognitive dissonance, nor the used instrument was a valid indicator. Furthermore, the visual processing in brain wave signals between reading interesting or not-interesting are similar [2], which further indicates that the web-logging or blink data are insufficient to detect the confirmation bias. Thus, other measuring techniques should be proposed to overcome this gap. Regarding cognitive dissonance, Boonprakong et al. [6] and Ploger et al. [24] found increased skin conductance levels on the dissenting stimulus, which suggests dissonance arousal. These works highlight the potential of using physiological signals to detect bias.

2.2 Affect and Cognition in Search Model

Emotion has been a topic of interest in IR for decades. In the 2000s, Kuhlthau [14] (2005) proposed the Information Search Process (ISP) model, which understands the search behaviors from three realms (cognitive, affective, and physical). The ISP model emphasized that all affective, cognitive and physical activities change throughout the search process. The ISP deconstructs the search process into six stages, initiation, selection, exploration, formulation, collection, and presentation. The key which causes the flows is due to the changing of *the feeling of uncertainty*.

“Uncertainty is a cognitive state that commonly causes affective symptoms of anxiety and lack of confidence.”

This concept is also related to the Anomalous States of Knowledge (ASK) model [4]. Later, Nahl [21] (2007) proposed the Social-Biological Information Technology model, which framed the interaction with technology (search system) in terms of sensorimotor, affect, and cognitive processes. This model interprets the search process as a sequence of the functionality of sensorimotor and the changing of affect and cognition which caused by the goal change at each search stage. The similarity of these two models further highlights the importance of cognitive and affective factors in understanding information-search behaviors, and the association with the appraisal theory and the orienting responses [26].

More recently, the fMRI study of Moshfeghi et al. [20] found the activated brain areas and corresponding brain activities differ in the transitions of search stages. The results showed that mainly motor processes were evoked for expressing the query in Query Formulation. In Query Submission, there were efforts to maintain the task and affective evaluation. The affective evaluation refers to the expectation of success and evaluation of the results of a query. In Relevance Judgment, the brain activities were highly activated for dynamic control of cognitive processes, examining the document and possibly leading to self-referential thought (which is also related to the affective state). Shovon et al. [27] analyzed the EEG signals of web searching. The query formulation sections (including typing) were found to have more intensive signals than the result list viewing section and content reading section. The author explained this reaction was because more sub-processes in cognition were encountered during query formulation. On the other hand, in most literature, search is referred to as a ‘goal-directed’ or ‘reward-seeking’ task [14, 19–21]. The cognitive processes and emotions during goal achievement can also be applied in the search process. Table 1 summarizes the flow of internal activities in different search stages.

In summary, as cognitive and affective activities are different in each search stage, they are also expected to evoke different orienting responses [26]. Following this trend, whether it is worth investing resources in the evaluation by digging and dissecting the underlying affect or cognitive processes in search is still a question.

3 METHODOLOGY

The ultimate aim is to quantify and measure confirmation bias (CB) in information retrieval (IR) using sensor data. There are four controlled user studies conducted, corresponding to each research question, gradually to achieve this goal. The first aims to look at the sensor data in a general information processing context. The second moves toward the specific IR context and investigates the efficiency of the sensor data applied in the search process. The third aims to investigate whether the sensor data are different in a binary stance scenario. The fourth aims to detect CB in the whole search process, including result selection and evaluation.

3.1 RQ1: Which Sensor Data can Discriminate the Tonic Information Processing Activities?

Information processing comprises two fundamental activities, information input and output. Both activities have two modalities, i.e., input via visuals (READ) or audio (LISTEN), and output via motion (WRITE or typing when using computers) or speech (SPEAK). The

Activity\Search Stage	Realization of Information Need	Query Formulation		Relevance Judgment (for highly relevant)
		Formulation	Submission	
Neuro [19]	dynamic restructuring of cognitive resources across the subtasks, maintain task set, switch attentions	mainly motor processes evoked for expressing the query in mind	maintain the task and affective evaluation (or reward-seeking)	dynamic control of cognitive processes, examining the document and possibly leading to self-referential thought (which is also related to the affective state)
Cognition [14]	Vague	focus ↑, more dynamic allocation of cognitive resources (cognitive activities proximity ↑)		focus ↑
Affect [4, 14]	goal-setting(feel motivated or the desire to do something [21])	—		value-attaching
	uncertainty ↑, confusion ↑, frustration, anxiety	confidence ↑, clarity, uncertainty ↓		confident ↑, interest ↑, satisfaction
	arousal ↑, valence ↓	arousal ↑, valence ↓		arousal ↓, valence ↑

Table 1: Neuro, Cognition, Affect changes in the information search process.

cognitive states and motor activities vary in four essential activities – READ, LISTEN, SPEAK, and WRITE. For example, all activities involve various cognitive states; SPEAK and WRITE also involve different motor activities. Hence, we anticipate they produce distinct physiological signals [11].

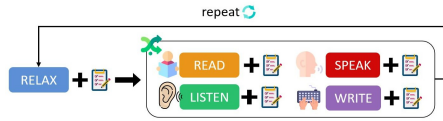


Figure 1: Study 1 Procedure

Experiment. The participant first needs to finish a background survey. The survey asked for sleeping hours prior and caffeine intake on the experiment day which might affect the participant’s cognition status. Figure 1 presents the experiment procedure. The experiment consisted of two sections in terms of complexity levels. Each section starts with a relaxing activity (RELAX), where the participant is asked to watch a relaxing video and minimal cognitive efforts are involved. Then the participant needs to complete four tasks, READ, LISTEN, SPEAK and WRITE. Specifically, the participant needs to read one article, listen to one article, answer a question by speaking, and answer a question by typing. After each task (including RELAX), the participants need to complete an engagement scale [23]. The instructor leaves the experiment room after the calibration and instruction to avoid interruption.

Results and Implications. We conducted a feasibility study and collected data from 7 participants (5M, 2F). All participants have at least professional working proficiency in English. The study received ethics approval from the author’s university, and the participants provided written consent before the experiment.

We compared the differences in physiological responses between 4 IPAs and a no-task baseline using trend analysis and observed some consistent trends that can discriminate IPAs. Our results confirm the potential of using sensor data to detect activities in information-processing contexts. Our analysis also suggests the need to use multi-modal data to analyze these activities. For example, gaze-related features can differentiate between the BASE and READ or BASE and WRITE, but adding BVP features can further differentiate READ and WRITE. Although data collected in this study are insufficient for a conclusive analysis of detecting IPAs, the results are sufficient to inform our study designs in subsequent research questions.

3.2 RQ2: How the Sensor Data Different in the Search Stages?

Based on the implications from Study 1, we have moved toward a more specific scenario, information retrieval. Due to the diversity of search systems, the 4 IPAs are considered as different interaction modalities with the system. Specifically, enter the search query by speaking or typing, or receive the search result by reading or listening.

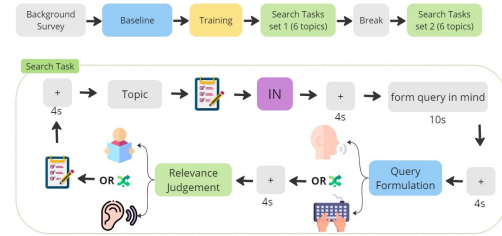


Figure 2: Study 2 Procedure

Experiment. To compare with the neuro-activities identified by Moshfeghi and Pollick [19], we design a similar setting and select the same search stages as theirs: Information Need (IN), Query Formulation (QF), Query Submission (QS), and Relevance Judgment (RJ). Note that as we are interested in different responses among search stages rather than particular evaluation criteria, RJ contains only highly relevant search results. In addition to the sensor used in Study 1, we also add the EEG sensor. EEG and pupil data can provide a robust metric for measuring cognitive activities; with them as a reference to the internal activities (e.g., cognition and emotion), we aim to investigate whether the peripheral data, e.g., EDA and BVP, can achieve a similar result.

After calibration and background survey, the participants are instructed to have an eyes-open section that collects the baseline data, followed by training. Each participant needs to complete the search tasks for 12 topics. A break time is provided in the middle. Each search task starts with a 4-second fixed-cross screen, followed by the topic. The participants need to rate their interests, familiarity, and perceived difficulty towards the topic. After that, a backstory to trigger the information need is presented. The participants then need to form a search query in mind within 10 seconds, followed by instructions about submitting the query by speaking (QF-S) or typing (QF-T). Next, the participants either read (RJ-R) or listen

(RJ-L) to a search result. In the end, the participants answer a judgment question (attention check) and rate perceived relevance and difficulty in understanding the search result. The 4-second fixed-cross screen is also shown after IN, QF, and QS stages to get clean signals. The sequences of topics and the combination of interaction modalities are randomized.

The topics and corresponding backstories are selected from the TREC2002-InformationNeed dataset [18]. To minimize risks of irrelevant emotions or cognitive bias, the topics are selected from the Understanding category which requires the participants to find information for understanding the topics. Besides, the topics related to crises, conspiracy, or politically sensitive are removed. For each topic, we manually select relevant articles from the TREC document collections, then use ChatGPT to generate a 150-word summary and a factual binary judgment question based on these articles.

Hypothesis. Based on the findings in Study 1 and Table 1, we have made the following hypothesis: **H1:** There are more subprocesses in QF than in other search stages; thus, the data can indicate cognitive activities can effectively detect QF. **H2:** Emotional responses elicited in RJ and IN are more diverse and intense than other search stages; thus, the data can indicate emotions can effectively detect RJ and IN. **a):** The negative emotions are mostly elicited in IN, due to the feeling of uncertainty [14]; thus, the data have decreasing trends for IN. **b):** When the search result is highly relevant, the positive emotions are mostly elicited in RJ, due to the feeling of uncertainty relieved [14]; thus, the data have increasing trends for RJ. **H3:** Features indicating motor activities can effectively detect interaction modalities. For example, BVP features are different between typing and speaking (higher in Speaking). Pupil dilates more in listening, and EDA decreases in reading.

3.3 RQ3: How the Sensor Data Describe Confirmation Bias?

The third study will investigate the CB under a binary stance context (supporting stance 1 or 2) in information evaluation. The experiment is modified based on Meppelink et al. [17] and Kessler and Zillich [12] (described in Section 2).

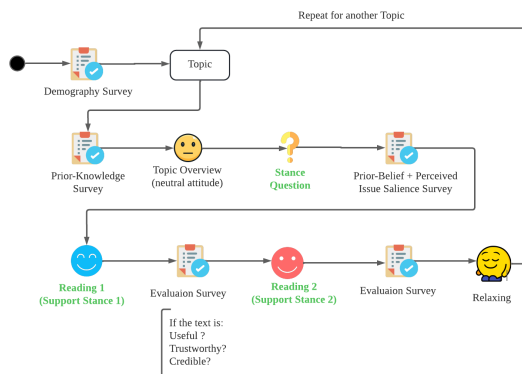


Figure 3: Study 3 Procedure

Experiment. As presented in Figure 3, after calibration and the background survey, the participants are instructed to collect the baseline data. Then, they read the task description. The aim of this study is covered by different aim, which is to understand people’s opinions about current events for educational purposes. The participants need to complete two topics. The order is randomized. For each topic, they first answer a prior-knowledge survey, then read a piece of overview information about the topic. The overview represents a neutral attitude. Then, the participants rate their prior beliefs and strength, and perceived issue salience about a stance question [12] (e.g., “Are you going to get the latest COVID-19 vaccination booster?”). The participants next read and evaluate a piece of information that supports Stance 1, and repeat for another equal-length information that supports Stance 2. The evaluation tasks are for assessing confirmation bias, as people tend to trust information that aligns with their beliefs over contradictory information [17]. A 5-minute break time is provided in the middle. The collected data includes EDA, BVP, gaze, head motion, and facial expressions.

3.4 RQ4: How can Sensor Data be Used to Measure and Quantify CB Levels in Information Seeking Processes?

After measuring CB in the simple information evaluation context, the fourth user study investigates the CB in a more complicated context, i.e., a complete IR process includes both information selection and evaluation. To stimulate the daily IR scenario, the experiment conditions will be more complicated but still be a controlled in-lab study to minimize noises in the sensors. It uses the sensor data which have been found useful in the two previous studies. The expected outcomes of this project are the analysis of the data changes in associate with the level of CB, and a machine-learning model to detect the level of CB using multi-modal data automatically. We also compare the proposed method (using sensor data) and other bias-quantifying methods (no sensor data), to further investigate the usability of using sensor data in this context.

4 POTENTIAL CONTRIBUTION

The final outcome of this project would be an approach to quantify and measure confirmation bias during the information retrieval process using multi-modal data. The approach includes the efficiency of data, in-depth analysis of the relationship between data and CB, important features extracted from the data, and machine learning models for measuring CB automatically. It also compares the efficiency of each sensor subset and discusses the practicality. Although the findings from Studies 1 and 2 could also contribute to voice interaction, the work only focuses on the visual-interacted information retrieval system due to the time limitation. A suggestion for a potential study can be using this quantifying approach as an index of the level of confirmation bias, and then evaluating the bias mitigation strategy. Another direction is to generalize this approach for more kinds of cognitive bias or other cognitive activities during the information-seeking process.

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